

Configure Route 53 Health Checks with Weighted Routing Policy

Summary:-

In this lab, you will configure Amazon Route 53 Health Checks for a Weighted Routing Policy. Health checks will be associated with multiple weighted records to continuously monitor the availability of EC2 web servers. You will simulate a server failure by blocking HTTP traffic, verify that Route 53 routes traffic only to healthy servers, restore the failed server, and confirm that traffic distribution resumes according to the configured weights.

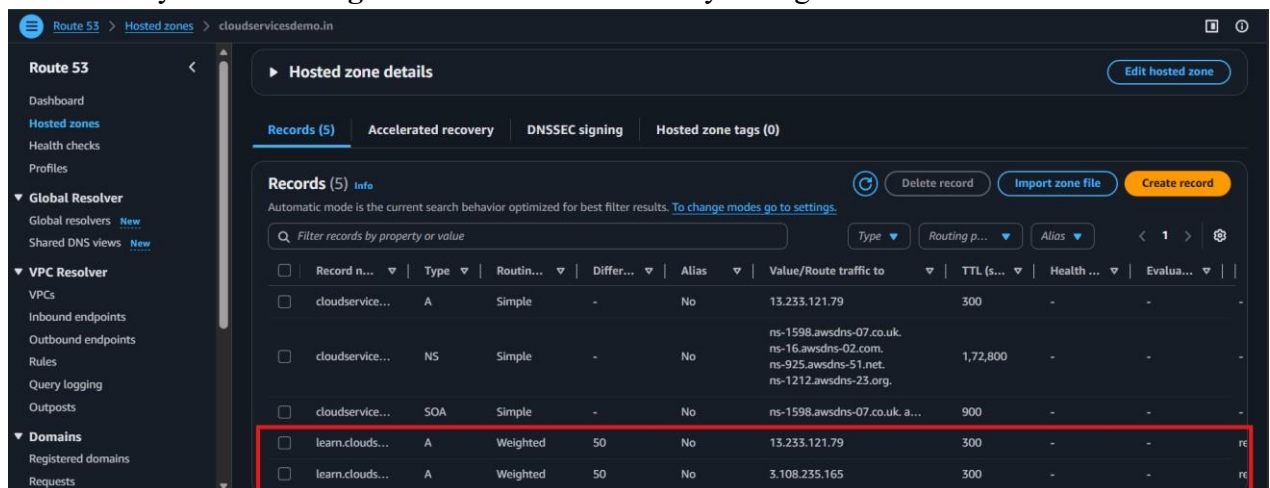
Let's Start:-

- Configure Route 53 Health Checks for weighted records.
- Associate Health Checks with EC2 web servers.
- Simulate a web server failure.
- Verify DNS routing based on Health Check status.
- Restore the failed server and verify normal traffic distribution.

Lab Steps:-

Step 1 – Verify the Existing Weighted Routing Policy

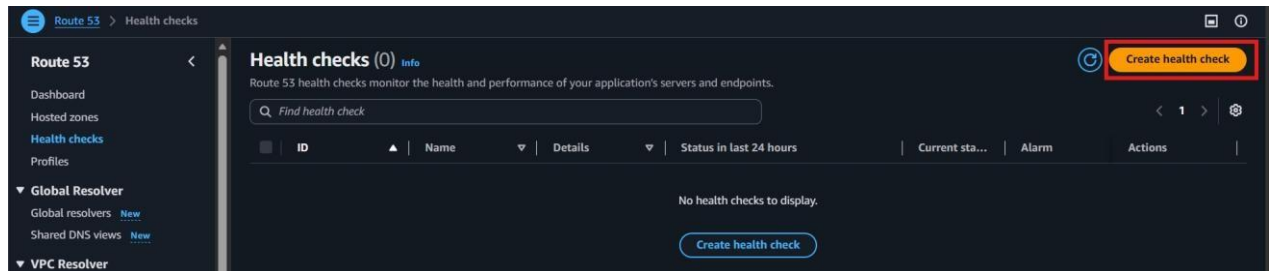
- Open the **Amazon Route 53** console.
- Select the existing Public Hosted Zone.
- Verify that two **Weighted A Records** are already configured.



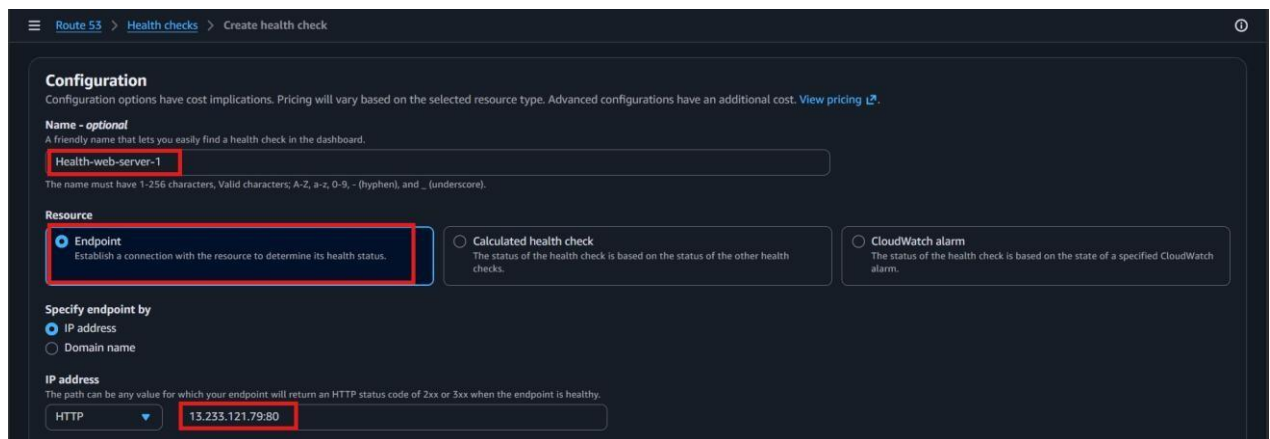
- Confirm that both EC2 web servers are accessible.
- **Note:** Before attempting this lab, complete **Steps 1–2** of the **Route53_DNS_Record_Types_AWS** lab to connect the website with the instance.

Step 2 – Create a Health Check for Web Server 1

- In the Route 53 console, select **Health Checks**.



- Click **Create Health Check**.
- Enter the following:
 - **Name:** Health-web-server-1
 - **Protocol:** HTTP
 - **IP Address:** Public IPv4 address of Web Server 1.
 - **Port:** 80



- Leave the remaining settings as default.
- Click **Create Health Check**.

Step 3 – Create a Health Check for Web Server 2

- Click **Create Health Check**.
- Enter the following:
 - **Name:** Health-web-server-2
 - **Protocol:** HTTP
 - **IP Address:** Public IPv4 address of Web Server 2
 - **Port:** 80

Route 53 > Health checks > Create health check

Configuration

Configuration options have cost implications. Pricing will vary based on the selected resource type. Advanced configurations have an additional cost. [View pricing](#)

Name - optional
A friendly name that lets you easily find a health check in the dashboard.

Health-web-server-2

The name must have 1-256 characters. Valid characters: A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Resource

☒ **Endpoint**
Establish a connection with the resource to determine its health status.

☐ **Calculated health check**
The status of the health check is based on the status of the other health checks.

☐ **CloudWatch alarm**
The status of the health check is based on the state of a specified CloudWatch alarm.

Specify endpoint by

☒ **IP address**

☐ **Domain name**

IP address
The path can be any value for which your endpoint will return an HTTP status code of 2xx or 3xx when the endpoint is healthy.

HTTP 3.108.235.165:80

▶ **Advanced configuration**

- Leave the remaining settings as default.
- Click **Create Health Check**.
- Wait 2–3 minutes until both Health Checks show the **Healthy** status.

Step 4 – Associate Health Checks with Weighted Records

- Open the **Hosted Zone**.
- Edit the weighted record for **Web Server 1**.

Route 53 > Hosted zones > cloudservicesdemo.in

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles
- ▼ **Global Resolver**
 - Global resolvers
 - Shared DNS views
- ▼ **VPC Resolver**
 - VPCs
 - Inbound endpoints
 - Outbound endpoints
 - Rules
 - Query logging
 - Outposts
- ▼ **Domains**
 - Registered domains

Records (5) Accelerated recovery DNSSEC signing Hosted zone tags (0)

Records (1/5) [Info](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias < 1 > ⚙

Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to
☐ cloudservi...	A	Simple	-	No	13.233.121.79
☐ cloudservi...	NS	Simple	-	No	ns-1598.awsdns-07.co.uk. ns-16.awsdns-02.com. ns-925.awsdns-51.net. ns-1212.awsdns-23.org.
☐ cloudservi...	SOA	Simple	-	No	ns-1598.awsdns-07.co.uk. a...
☒ learn.clou...	A	Weighted	50	No	13.233.121.79
☐ learn.clou...	A	Weighted	50	No	3.108.235.165

Record details

[Edit record](#)

Record name
learn.cloudservicesdemo.in

Record type
A

Value
13.233.121.79

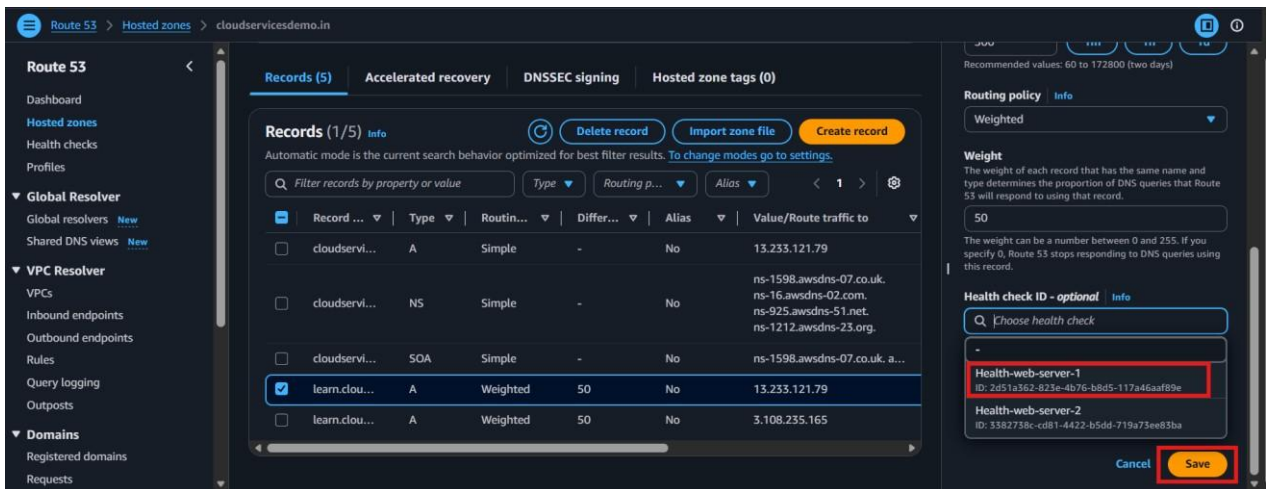
Alias
No

TTL (seconds)
300

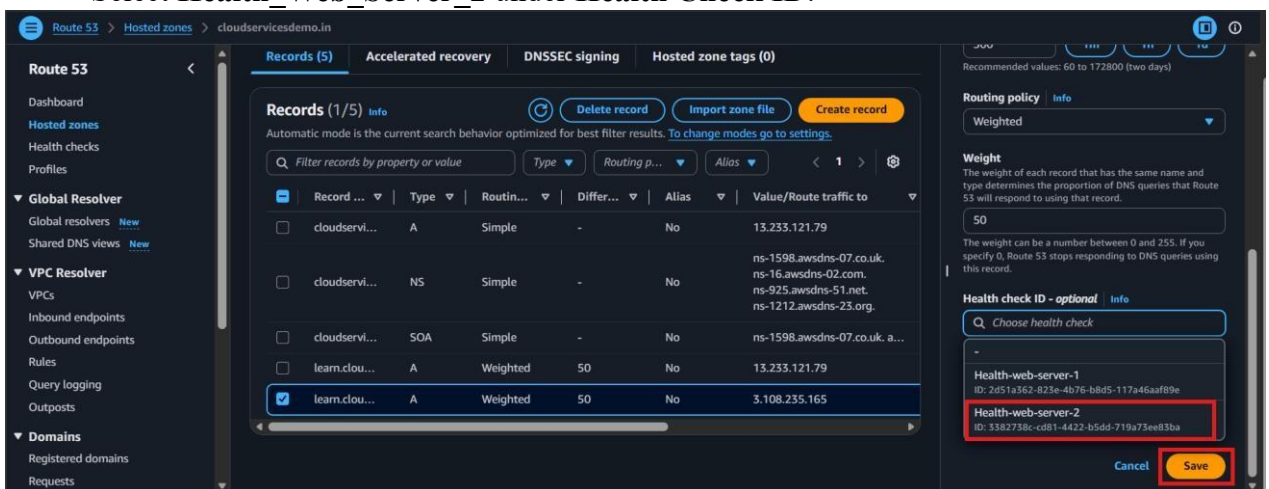
Routing policy
Weighted

Weight
50

- Select **Health_Web_Server_1** under **Health Check ID**.



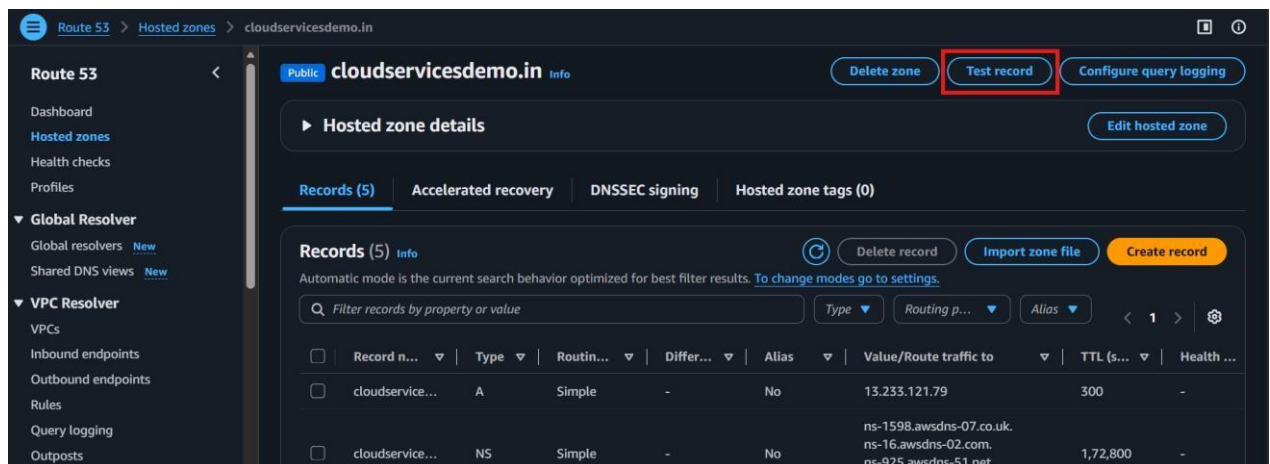
- Save the record.
- Edit the weighted record for **Web Server 2**.
- Select **Health_Web_Server_2** under **Health Check ID**.



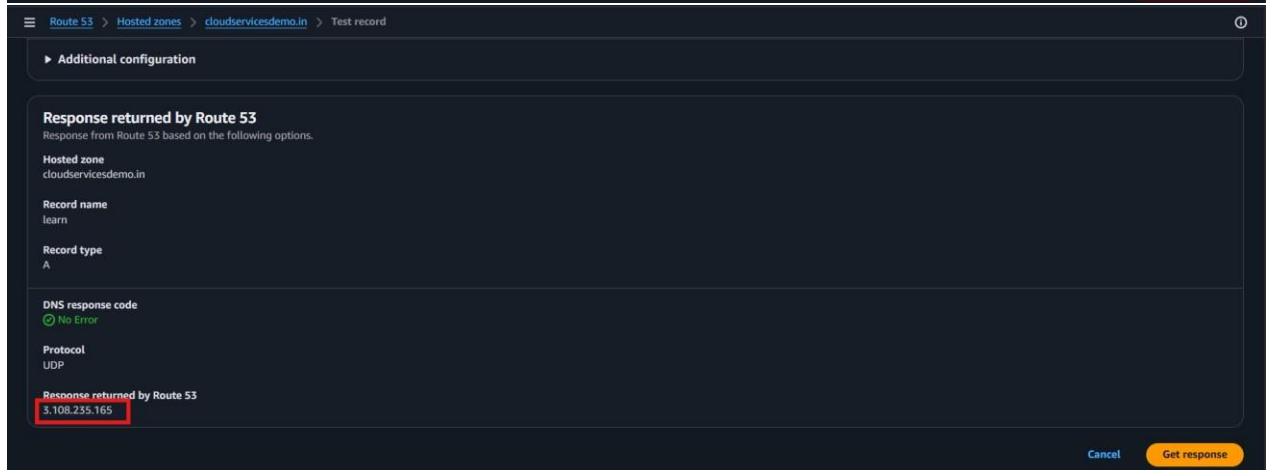
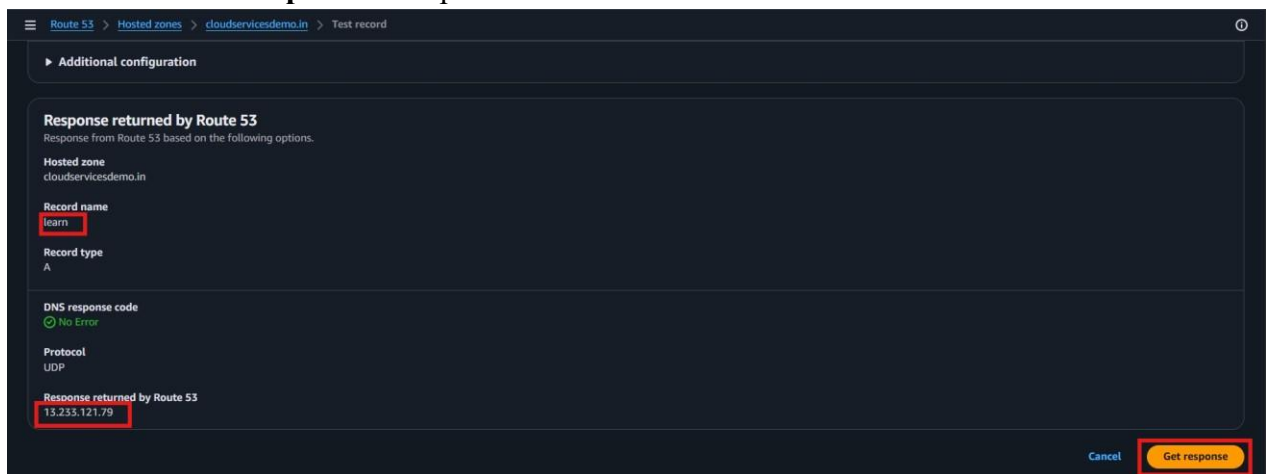
- Save the record.

Step 5 – Verify Normal DNS Resolution

- Open the **Hosted Zone**.
- Create a **Test Record** for learn.



- Click Get Response multiple times.

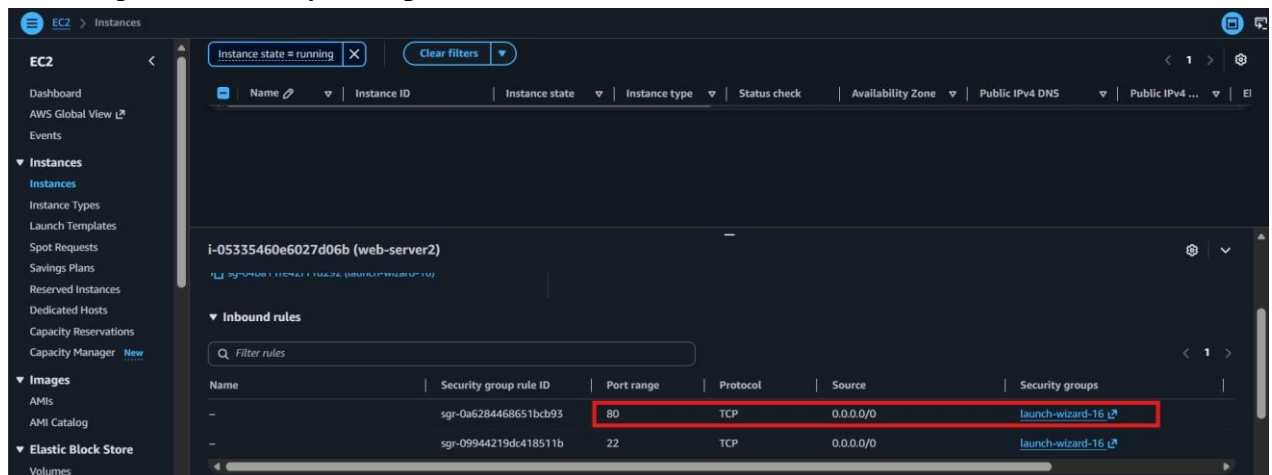


- Verify that both web server IP addresses are returned.

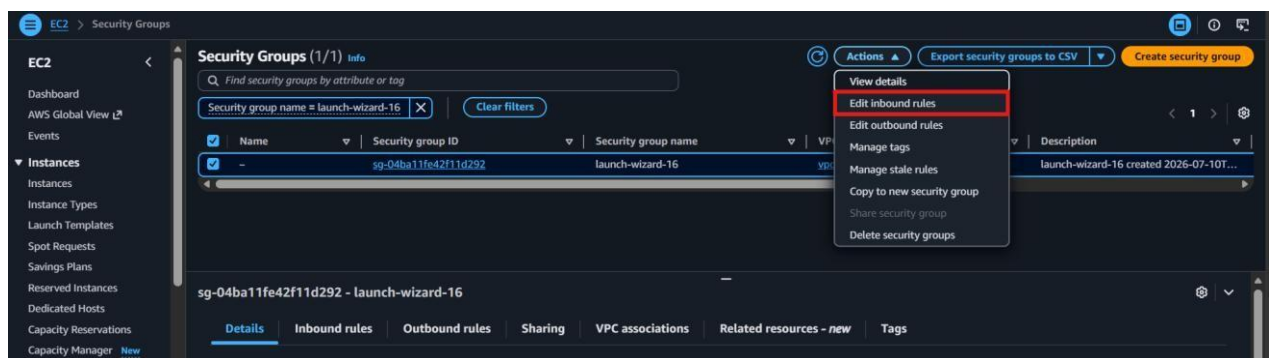
Step 6 – Simulate a Web Server Failure

- Open the EC2 console.
- Select Web Server 2(any).

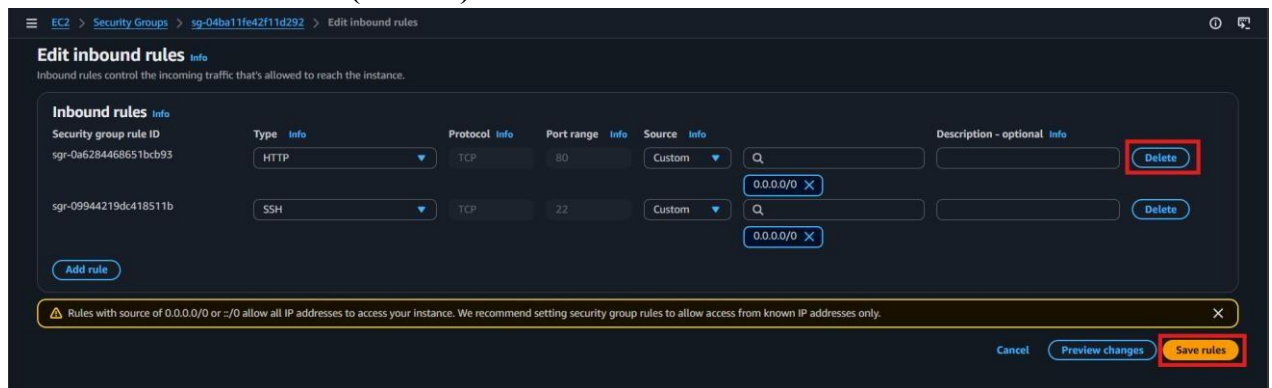
- Open its Security Group.



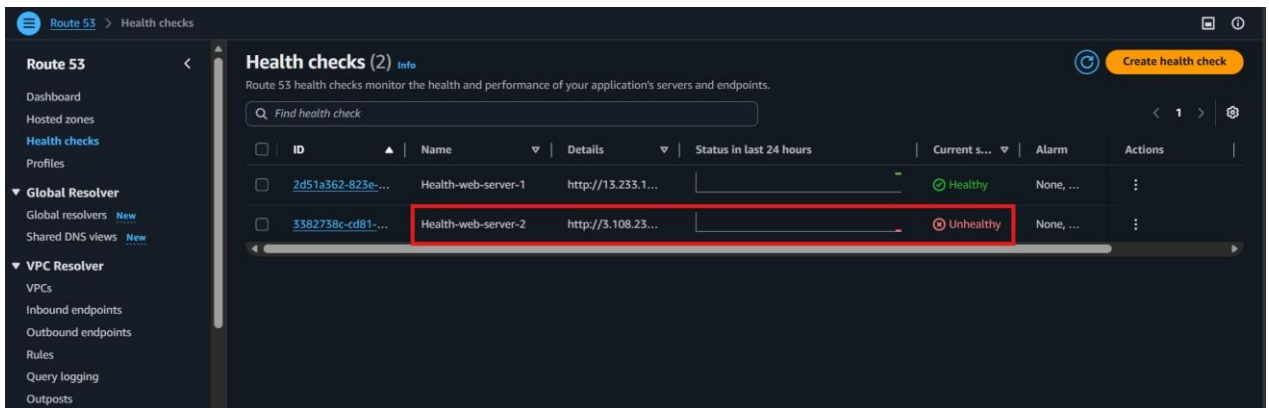
- Edit the inbound rules.



- Remove the HTTP (Port 80) rule.



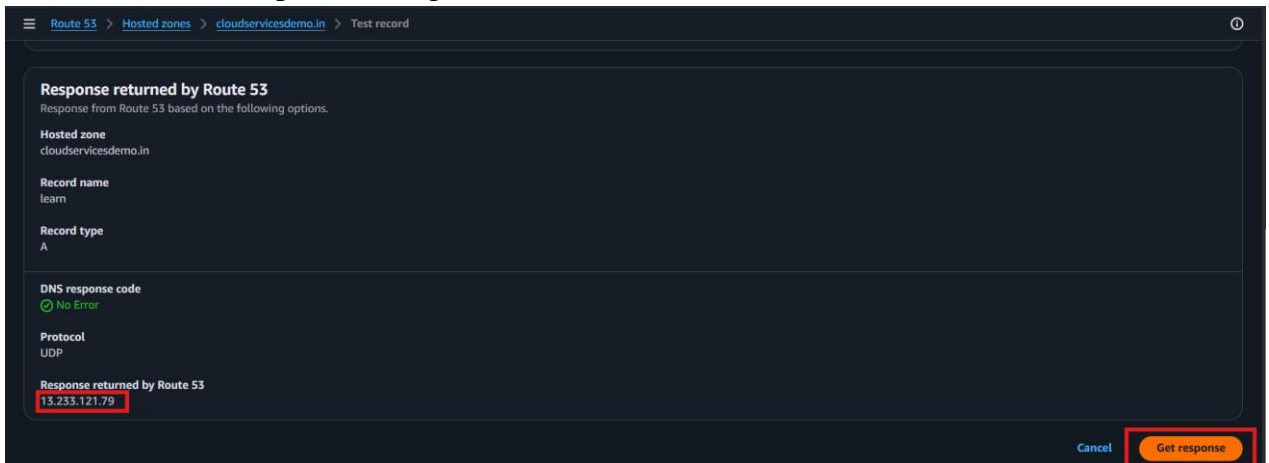
- Save the changes.
- Verify that the website hosted on Web Server 2 is no longer accessible.



- Wait until the corresponding Health Check status changes to **Unhealthy**.

Step 7 – Verify DNS Routing

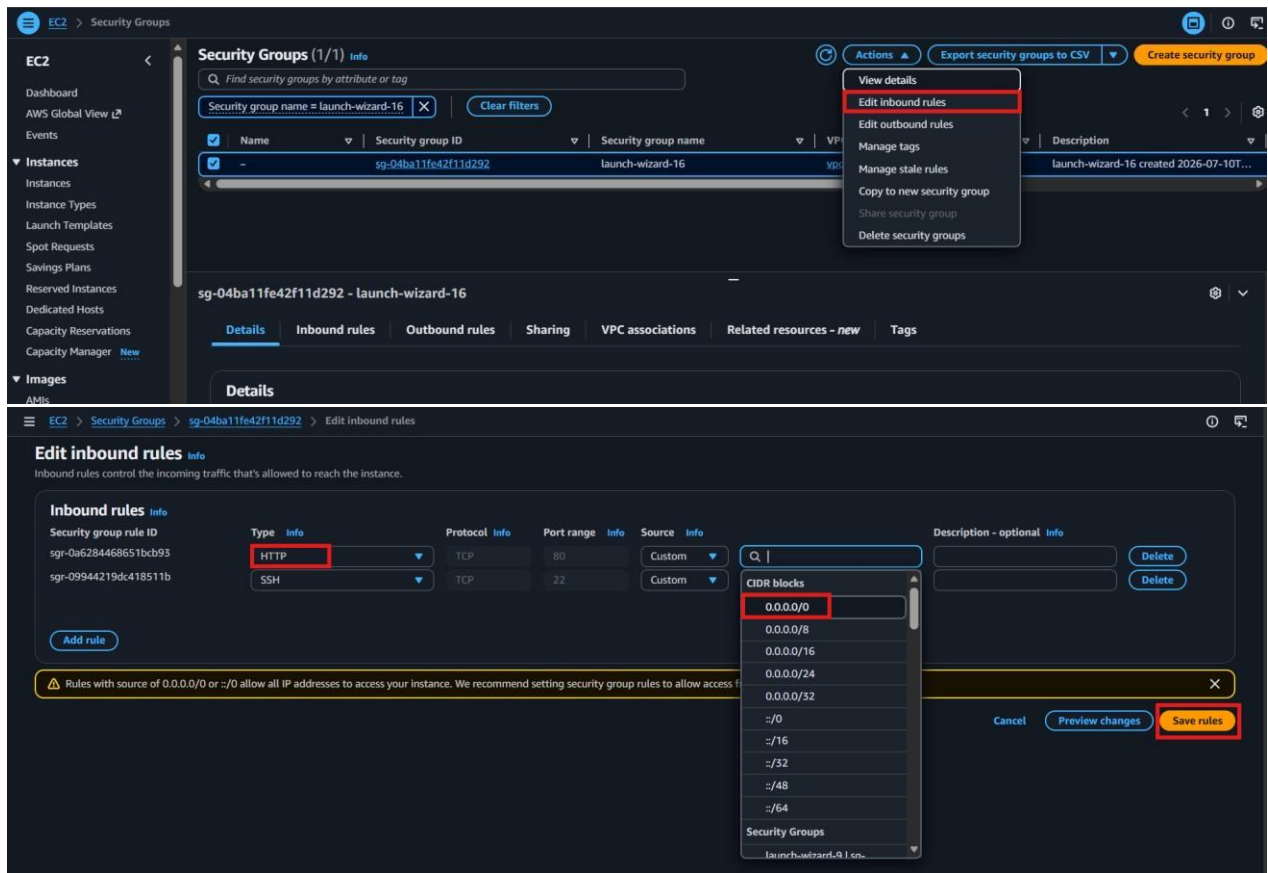
- Open **Test Record** in the **Hosted Zone**.
- Enter **learn**.
- Click **Get Response** multiple times.



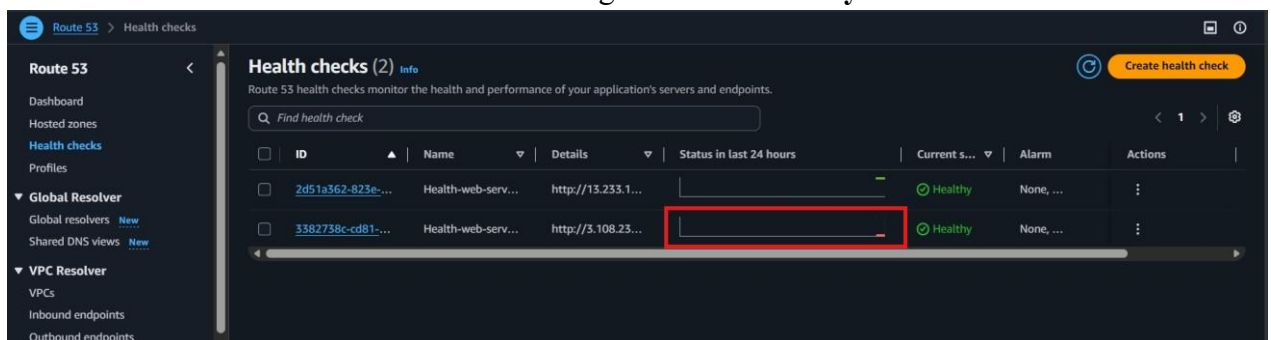
- Verify that only **Web Server 1** is returned.

Step 8 – Restore the Web Server

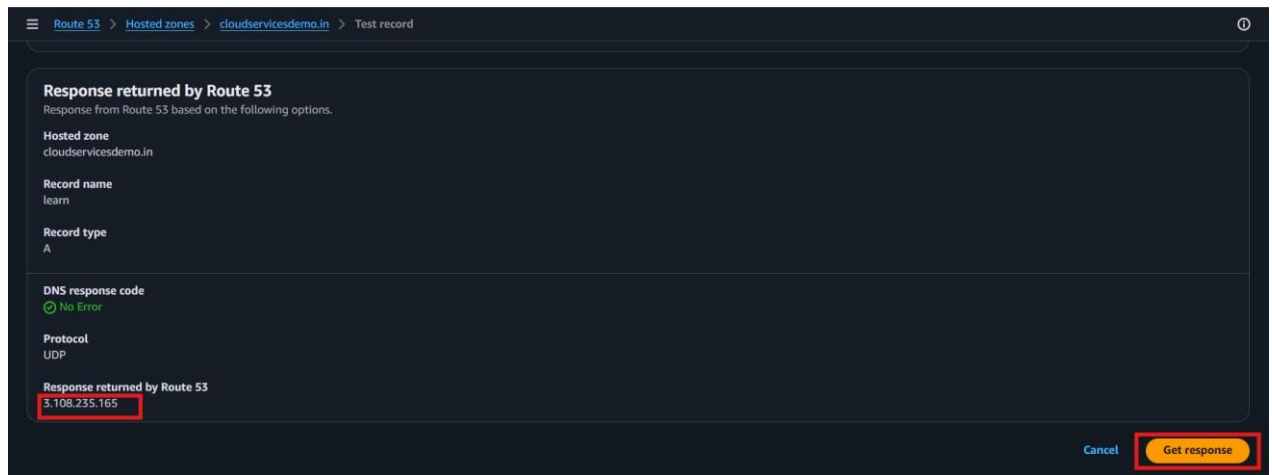
- Edit the Security Group of **Web Server 2**.
- Add the **HTTP (Port 80)** inbound rule.



- Save the changes.
- Wait until the Health Check status changes back to **Healthy**.



- Open **Test Record** in the Hosted Zone, **enter** learn, and click **Get Response** multiple times.



- Verify that Route 53 again resolves the IP addresses of both web servers.

Verification

- Successfully created Route 53 Health Checks.
- Associated Health Checks with Weighted Routing Policy records.
- Simulated a web server failure successfully.
- Verified that Route 53 routed traffic only to healthy web servers.
- Restored the failed server and confirmed normal weighted traffic distribution.